

Лабораторная работа №15

Администрирование локальных сетей

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Информация

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- <https://github.com/chelibos?tab=repositories>

Настроить динамическую маршрутизацию между территориями организации.

Выполнение лабораторной работы

Настроиваю динамическую маршрутизацию по протоколу OSPF

```
msk-donskaya-aodikach-gw-01>enable
Password:
msk-donskaya-aodikach-gw-01#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-aodikach-gw-01(config)#router ospf 1
msk-donskaya-aodikach-gw-01(config-router)#router-id 10.128.254.1
msk-donskaya-aodikach-gw-01(config-router)#network 10.0.0.0 0.255.255.255 area 0
msk-donskaya-aodikach-gw-01(config-router)#exit
msk-donskaya-aodikach-gw-01(config)#exit
msk-donskaya-aodikach-gw-01#
```

Рис. 1: Настройка маршрутизатора msk-donskaya

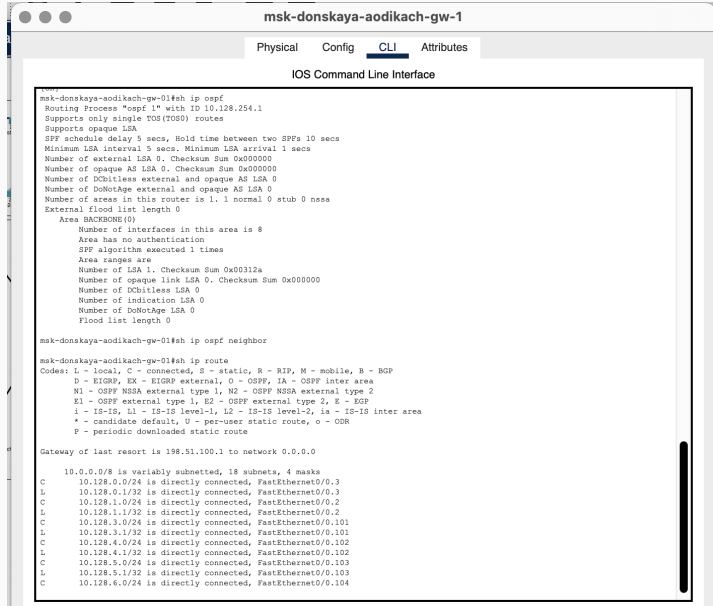


Рис. 2: Проверка состояния протокола OSPF на маршрутизаторе

```

msk-q42-aodikach-gw-1>enable
Password:
msk-q42-aodikach-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-aodikach-gw-1(config)#router ospf 1
msk-q42-aodikach-gw-1(config-router)#router-id 10.128.254.2
msk-q42-aodikach-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
msk-q42-aodikach-gw-1(config-router)#exit
msk-q42-aodikach-gw-1(config)#exit
msk-q42-aodikach-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-aodikach-gw-1#wr m
Building configuration...
[OK]
msk-q42-aodikach-gw-1#
00:09:47: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.3 on FastEthernet1/0.202 from LOADING to FULL, Loading Done

msk-q42-aodikach-gw-1#sh ip ospf
Routing Process "ospf 1" with ID 10.128.254.2
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 3
    Area has no authentication
    SPF algorithm executed 2 times
    Area ranges are
    Number of LSA 3. Checksum Sum 0x01779e
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0

```

Рис. 3: Настройка и проверка протокола

```

msk-hostel-aodikach-gw-1>enable
Password:
msk-hostel-aodikach-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-hostel-aodikach-gw-1(config)#router ospf 1
msk-hostel-aodikach-gw-1(config-router)#router-id 10.128.254.3
msk-hostel-aodikach-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
msk-hostel-aodikach-gw-1(config-router)#exit
msk-hostel-aodikach-gw-1(config)#exit
msk-hostel-aodikach-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-hostel-aodikach-gw-1#wr m
Building configuration...
[OK]
msk-hostel-aodikach-gw-1#
00:09:47: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.2 on Vlan202 from LOADING to FULL, Loading Done

msk-hostel-aodikach-gw-1#sh ip ospf
Routing Process "ospf 1" with ID 10.128.254.3
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 2
    Area has no authentication
    SPF algorithm executed 2 times
    Area ranges are
    Number of LSA 3. Checksum Sum 0x01779e
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0

```

Рис. 4: Настройка и проверка протокола

```

passw...
sch-sochi-aodikach-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-aodikach-gw-1(config)#
sch-sochi-aodikach-gw-1(config)#router ospf 1
sch-sochi-aodikach-gw-1(config-router)#router-id 10.128.254.4
sch-sochi-aodikach-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
sch-sochi-aodikach-gw-1(config-router)#exit
sch-sochi-aodikach-gw-1(config)#exit
sch-sochi-aodikach-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-aodikach-gw-1#wr m
Building configuration...
[OK]
sch-sochi-aodikach-gw-1#sh ip ospf
Routing Process "ospf 1" with ID 10.128.254.4
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
Area BACKBONE(0)
Number of interfaces in this area is 3
Area has no authentication
SPF algorithm executed 1 times
Area ranges are
Number of LSA 1. Checksum Sum 0x000f60
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0
sch-sochi-aodikach-gw-1#

```

Рис. 5: Настройка и проверка протокола

Настраиваю связь сети 42 квартала в Москве с сетью филиала в г. Сочи

```
provider-aodikach-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
provider-aodikach-sw-1(config)#vlan 7
provider-aodikach-sw-1(config-vlan)#name q42-sochi
provider-aodikach-sw-1(config-vlan)#exit
provider-aodikach-sw-1(config)#interface vlan7
provider-aodikach-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up

provider-aodikach-sw-1(config-if)#no shutdown
      ^
% Invalid input detected at '^' marker.

provider-aodikach-sw-1(config-if)#no shutdown
provider-aodikach-sw-1(config-if)#exit
provider-aodikach-sw-1(config)#
provider-aodikach-sw-1(config)#exit
provider-aodikach-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-aodikach-sw-1#wr m
Building configuration...
[OK]
```

Рис. 6: Настройка коммутатора

```
msk-q42-aodikach-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-aodikach-gw-1(config)#interface f0/1.7
msk-q42-aodikach-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/1.7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.7, changed state to up

msk-q42-aodikach-gw-1(config-subif)#encapsulation dot1Q7
                                     ^
% Invalid input detected at '^' marker.

msk-q42-aodikach-gw-1(config-subif)#encapsulation dot1Q 7
msk-q42-aodikach-gw-1(config-subif)#ip address 10.128.255.9 255.255.255.252
msk-q42-aodikach-gw-1(config-subif)#description sochi
msk-q42-aodikach-gw-1(config-subif)#exit
msk-q42-aodikach-gw-1(config)#
```

Рис. 7: Настройка маршрутизатора

```
sch-sochi-aodikach-sw-1>enable
Password:
sch-sochi-aodikach-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-aodikach-sw-1(config)#vlan 7
sch-sochi-aodikach-sw-1(config-vlan)#name q42-sochi
sch-sochi-aodikach-sw-1(config-vlan)#exit
sch-sochi-aodikach-sw-1(config)#interface vlan7
sch-sochi-aodikach-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up

sch-sochi-aodikach-sw-1(config-if)#no shutdown
sch-sochi-aodikach-sw-1(config-if)#exit
sch-sochi-aodikach-sw-1(config)#
```

Рис. 8: Настройка коммутатора

```
sch-sochi-aodikach-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-aodikach-gw-1(config)#interface f0/0.7
sch-sochi-aodikach-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.7, changed state to up

sch-sochi-aodikach-gw-1(config-subif)#encapsulation dot1Q 7
sch-sochi-aodikach-gw-1(config-subif)#ip address 10.128.255.10 255.255.255.252
sch-sochi-aodikach-gw-1(config-subif)#description q42
sch-sochi-aodikach-gw-1(config-subif)#exit
sch-sochi-aodikach-gw-1(config)#exit
sch-sochi-aodikach-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-aodikach-gw-1#wr m
Building configuration...
[OK]
sch-sochi-aodikach-gw-1#
```

Рис. 9: Настройка маршрутизатора

Просматриваю движение ICMP пакета

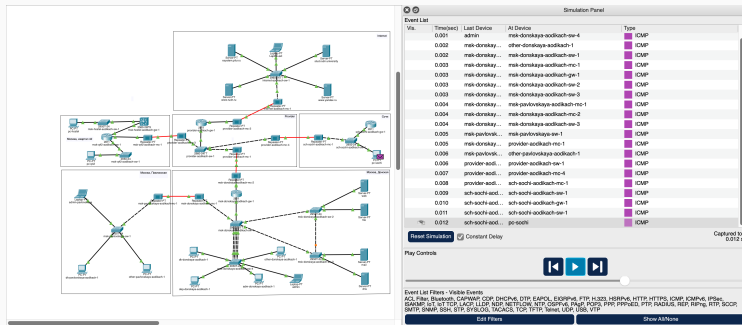


Рис. 10: Просмотр симуляции

Отключаю vlan 6 на провайдере и заново запускаю симуляцию

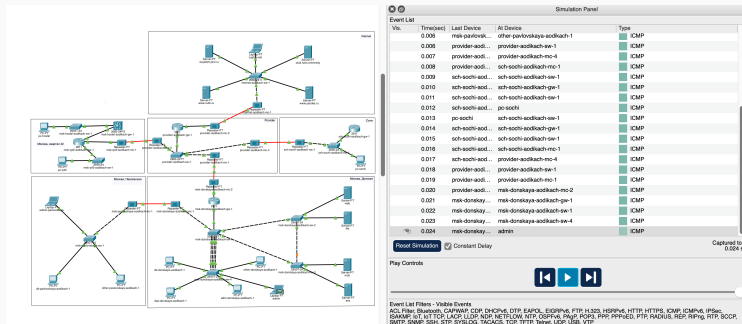


Рис. 11: Просмотр симуляции

Восстанавливаю работу vlan 6 на устройстве провайдера

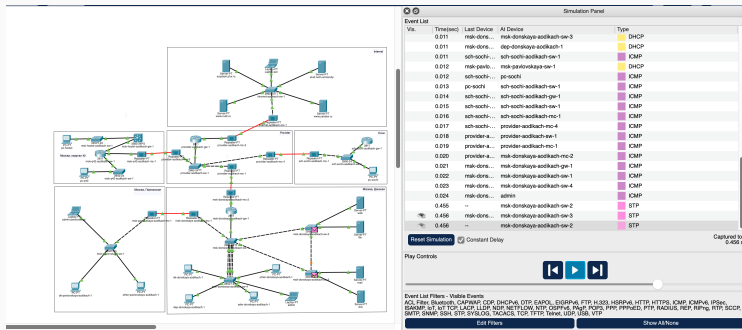


Рис. 12: Просмотр симуляции

Настроила динамическую маршрутизацию между территориями организации.